

It is also possible to set up a Firefox OS emulator since the OS is open source. This will actually run Firefox OS in a virtual mobile phone just like the Android emulator does. In the case of Firefox OS, where the app is essentially a web application, doing so is of dubious value. The performance of neither Simulator nor Emulator is comparable to that of a real phone, and the simulator is just quicker and easier to work with.

Examples for better understanding

Let's illustrate the difference between emulation and simulation is with a final few pointed examples. QEMU is a machine emulator, it is used by the Android SDK — among many other software — to emulate a mobile phone to run Android on. On the other hand you have a software like SPICE (Simulation Program with Integrated Circuit Emphasis). SPICE simulates electronic circuits allowing you to study how an electronic system behaves without needing to create an actual physical circuit. A circuit built in SPICE however, cannot replace an actual physical circuit! One cannot create a simulated circuit board in SPICE and have it interact with a real electronic component if one wants to test it.

A QEMU machine used by the Android emulator can run the actual code that runs on an actual mobile phone. It is emulating the mobile phone and as such can replace a real mobile phone. The same cannot be said of a simulation. The exact binary that is running in the iOS simulator will not run in the actual iOS device, the SDK creates a different package for the actual phone. Also, Train simulator might give you all the pleasant ambiance and visual splendour of driving a train through Europe, but it cannot replace an actual train ride, you will not actually be transported anywhere, other than perhaps in your mind.

As you can see these three concepts while quite interlinked and often overlapping, are not the same thing, and often do not serve similar purposes. In a colloquial setting however, there is often enough leeway to confuse one for another.

There is no point in being purposely stringent in enforcing terms that are already in popular use in different ways. Look at poor Wine (Wine is not an Emulator). It is a software that has named itself after the fact that it is not an emulator, yet in the public consciousness it is something that emulates Windows so you can run Windows applications on Linux.

Is it accurate to talk about mobile phone emulators and include iOS simulator, and Firefox OS Simulator in it? Should we call them “Software